

# Reverse Ultimatum Game with Exogenous Breakdown Risk

GitHub repository: <https://github.com/JonSnow016/Reverse-Ultimatum-Game-Simulation.git>

## Overview

This thesis analyses bargaining under exogenous negotiation breakdown, where talks may terminate after a rejection for reasons beyond either party's control. Rather than modelling delay through fixed deadlines or deterministic waiting costs, it introduces a stochastic collapse risk that makes the continuation value fragile and strategically consequential. This structure captures a common feature of real negotiations in wage setting, contracting, and political agreement formation, where external shocks and institutional constraints can abruptly end discussions. By formalizing how breakdown risk reshapes acceptance thresholds and optimal offers, the thesis provides a tractable foundation for studying learning and behavior of autonomous negotiating agents in environments with probabilistic failure.

## Model and theoretical benchmark

The setting is a Reverse Ultimatum Game with a fixed pie of 10 tokens. A proposer makes an offer. If the responder rejects, the proposer may increase the offer, but each rejection carries an exogenous breakdown risk  $p$  that ends the negotiation and yields zero to both sides. Using game theoretical solution of backward induction, the thesis derives a stable acceptance threshold for the responder and shows that a proposer has an incentive to offer the smallest amount that clears this threshold immediately, leading to agreement in the first stage. Higher breakdown risk predicts earlier acceptance and a shift in bargaining power toward the proposer.

## Learning and simulation evidence

The theoretical logic is tested with reinforcement learning. Populations of Q learning agents repeatedly play the game and update strategies based on realised payoffs. Two variants are studied: a constrained version where offers can only increase by one token after rejection, and an unconstrained version where proposers can jump to any higher offer. In both cases, agents converge to equilibrium like behaviour: responders learn a consistent acceptance cutoff and proposers learn to start near that cutoff, producing stable early agreement patterns across values of  $p$ .

## Reputation and evolutionary dynamics

The thesis extends the framework with heterogeneity and selection. Responders can differ in their acceptance thresholds, and proposers observe a short history of each responder's past decisions. Over generations, tougher responders can build a reputation that induces higher initial offers. These strategies can become dominant even when breakdown risk is present, illustrating how history and perceived commitment can move outcomes away from the standard equilibrium benchmark.

## Future Research

Future research should examine (i) endogenous and time varying breakdown risk, including asymmetric information about  $p$ , to identify how perceived collapse risk shapes acceptance thresholds and offer paths; (ii) richer learning and opponent modelling beyond tabular Q learning to test the robustness of equilibrium convergence under generalisation and non stationarity; and (iii) empirical calibration and validation using laboratory or field bargaining data.